

Statistical Theory and Modeling (ST2601)

Lecture 8 - Linear regression in vector form

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Overview

- Vectors and matrices - minimal intro to linear algebra
- Linear regression in vector form
- Multivariate normal distribution

Goals of the lecture

- **Linear regression in vector form**

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

- **Least squares estimate** of regression coefficients

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

- **Multivariate normal distribution** $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with pdf

$$f(\mathbf{x}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

- What's the deal with all the bold letters? **Linear algebra**.

- Worth the trip. Very useful for linear regression and more.

Vectors

- Linear algebra: a **vector** is an object containing **real numbers**

$$\mathbf{a} = \begin{pmatrix} 1 \\ 3 \\ 5 \\ 3 \end{pmatrix}$$

- Common default: a vector is a **column vector**.
- The **transpose** of a vector is a row vector

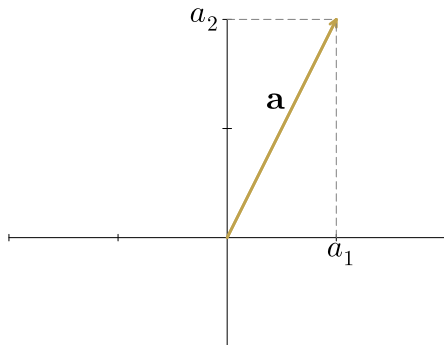
$$\mathbf{a}^T = (1 \quad 3 \quad 5 \quad 3)$$

- R:
 - > a = c(1,2,5,3)
 - > t(a) # transpose

Visualizing vectors in 2D

- **2D vector**. Directed line (arrow) in $\mathbb{R}^2$.

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

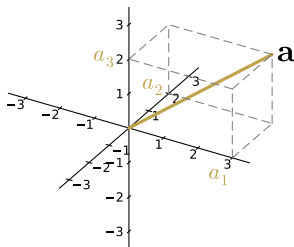


Visualizing vectors in 3D

- **3D vector.** Directed line (arrow) in $\mathbb{R}^3$.

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$$

$$\mathbf{a} = (3, 2, 2)^\top$$



Vector addition and subtraction

- **Adding two vectors** with the same number of elements

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \quad \mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{pmatrix}$$

- **Subtracting** a vector from another vector

$$\mathbf{a} - \mathbf{b} = \begin{pmatrix} a_1 - b_1 \\ a_2 - b_2 \\ a_3 - b_3 \end{pmatrix}$$

- Both these operations can be visualized geometrically.

Vector multiplication

- In R `a*b` will do **elementwise multiplication**

$$\mathbf{a} * \mathbf{b} = \begin{pmatrix} a_1 b_1 \\ a_2 b_2 \\ a_3 b_3 \end{pmatrix}$$

- In `a%*%b` will compute the **dot product**

$$\mathbf{a} \cdot \mathbf{b} = \mathbf{a}^\top \mathbf{b} = \begin{pmatrix} a_1 & a_2 & a_3 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

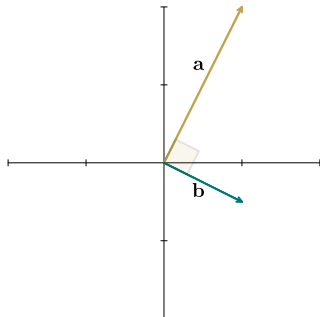
- In general the **dot product** is

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i$$

Orthogonal vectors

- Two vectors are **orthogonal** if their dot product is zero

$$\mathbf{a} \cdot \mathbf{b} = 0$$



- Example in 3D:

$$\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -6 \\ 2 \\ 2 \end{pmatrix}$$

Matrices

- A **matrix** is like a table, it has **rows** and **columns**

$$\mathbf{X} = \begin{pmatrix} 2 & 3 & 1 \\ 3 & 2 & 0 \end{pmatrix}$$

- This is a 2×3 matrix since it has 2 rows and 3 columns.
- View a $p \times q$ matrix as q column vector stacked horizontally

$$\mathbf{X} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_q \\ | & | & \cdots & | \end{pmatrix}$$

- Example: the following three vectors give the matrix above

$$\mathbf{x}_1 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \quad \mathbf{x}_2 = \begin{pmatrix} 3 \\ 2 \end{pmatrix}, \quad \mathbf{x}_3 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

- > `x1 = c(2,3); x2 = c(3,2); x3=c(1,0);`
- > `cbind(x1,x2,x3) # column bind. Also rbind exists`

Matrix-Vector multiplication

- $\mathbf{A}$ is an $m \times n$ matrix $\mathbf{A}$
- $\mathbf{b}$ is an n -element vector
- **Matrix-vector product**: dot product of each row in $\mathbf{A}$ with $\mathbf{b}$

$$\mathbf{A}_{(m \times n)} = \begin{pmatrix} - & \mathbf{a}_1^\top & - \\ - & \mathbf{a}_2^\top & - \\ & \vdots & \\ - & \mathbf{a}_m^\top & - \end{pmatrix} \quad \mathbf{b}_{(n \times 1)} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$$
$$\mathbf{A}\mathbf{b} = \begin{pmatrix} \mathbf{a}_1^\top \mathbf{b} \\ \mathbf{a}_2^\top \mathbf{b} \\ \vdots \\ \mathbf{a}_m^\top \mathbf{b} \end{pmatrix}$$

Matrix-Matrix multiplication

- Matrix product of $\mathbf{A}$ and $\mathbf{B}$: pairwise dot product of a row in $\mathbf{A}$ and a column in $\mathbf{B}$

$$\mathbf{A}_{(m \times n)} = \begin{pmatrix} - & \mathbf{a}_1^\top & - \\ - & \mathbf{a}_2^\top & - \\ & \vdots & \\ - & \mathbf{a}_m^\top & - \end{pmatrix} \quad \mathbf{B}_{(n \times r)} = \begin{pmatrix} | & | & \cdots & | \\ \mathbf{b}_1 & \mathbf{b}_2 & & \mathbf{b}_r \\ | & | & & | \end{pmatrix}$$

$$\mathbf{AB} = \begin{pmatrix} \mathbf{a}_1^\top \mathbf{b}_1 & \mathbf{a}_1^\top \mathbf{b}_2 & \cdots & \mathbf{a}_1^\top \mathbf{b}_r \\ \mathbf{a}_2^\top \mathbf{b}_1 & \mathbf{a}_2^\top \mathbf{b}_2 & \cdots & \mathbf{a}_2^\top \mathbf{b}_r \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{a}_m^\top \mathbf{b}_1 & \mathbf{a}_m^\top \mathbf{b}_2 & \cdots & \mathbf{a}_m^\top \mathbf{b}_r \end{pmatrix}$$

Matrix-Matrix multiplication

■ Example

$$\mathbf{A} = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$\begin{aligned} \mathbf{AB} &= \begin{pmatrix} (2 \ 3) \begin{pmatrix} 1 \\ 0 \end{pmatrix} & (2 \ 3) \begin{pmatrix} 2 \\ 1 \end{pmatrix} \\ (3 \ 2) \begin{pmatrix} 1 \\ 0 \end{pmatrix} & (3 \ 2) \begin{pmatrix} 2 \\ 1 \end{pmatrix} \end{pmatrix} \\ &= \begin{pmatrix} 2 \cdot 1 + 3 \cdot 0 = 2 & 2 \cdot 2 + 3 \cdot 1 = 7 \\ 3 \cdot 1 + 2 \cdot 0 = 3 & 3 \cdot 2 + 2 \cdot 1 = 8 \end{pmatrix} \end{aligned}$$

- > `A = matrix(c(2,3,3,2), 2, 2, byrow = TRUE)`
- > `B = matrix(c(1,2,0,1), 2, 2, byrow = TRUE)`
- > `A%%B` # `A*B` would do elementwise multiplication

Linear regression - one observation

- One observation

$$y = \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon$$

- In vector form

$$y = \underbrace{\begin{pmatrix} x_1 & \cdots & x_p \end{pmatrix}}_{\mathbf{x}^\top} \underbrace{\begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}}_{\boldsymbol{\beta}} + \varepsilon = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon$$

- Add a one for the **intercept**

$$\begin{pmatrix} 1 & x_1 & \cdots & x_p \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}$$

- The i th observation

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$$

Linear regression - all observations

- The i th observation

$$y_i = \mathbf{x}_i^\top \boldsymbol{\beta} + \varepsilon_i$$

- All $i = 1, \dots, n$ observations stacked under each other

$$\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \boldsymbol{\beta} \\ \mathbf{x}_2^\top \boldsymbol{\beta} \\ \vdots \\ \mathbf{x}_n^\top \boldsymbol{\beta} \end{pmatrix} + \begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}$$

- With matrix-vector multiplication

$$\begin{pmatrix} \mathbf{x}_1^\top \boldsymbol{\beta} \\ \mathbf{x}_2^\top \boldsymbol{\beta} \\ \vdots \\ \mathbf{x}_n^\top \boldsymbol{\beta} \end{pmatrix} = \underbrace{\begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix}}_{\mathbf{X}} \boldsymbol{\beta} = \mathbf{X} \boldsymbol{\beta}$$

- $\mathbf{X}$ is the $n \times p$ **covariate matrix** with n observations as rows.

Linear regression

Linear regression in vector form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

Least squares estimate = maximum likelihood estimate

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

We now understand that

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \sum_{i=1}^n x_{1i}^2 & \sum_{i=1}^n x_{1i}x_{2i} & \cdots & \sum_{i=1}^n x_{1i}x_{pi} \\ \sum_{i=1}^n x_{1i}x_{2i} & \sum_{i=1}^n x_{2i}^2 & \cdots & \sum_{i=1}^n x_{2i}x_{pi} \\ \vdots & \vdots & \ddots & \vdots \\ \sum_{i=1}^n x_{1i}x_{pi} & \sum_{i=1}^n x_{2i}x_{pi} & \cdots & \sum_{i=1}^n x_{pi}^2 \end{pmatrix}$$
$$\mathbf{X}^\top \mathbf{y} = \begin{pmatrix} \sum_{i=1}^n x_{1i}y_i \\ \sum_{i=1}^n x_{2i}y_i \\ \vdots \\ \sum_{i=1}^n x_{pi}y_i \end{pmatrix}$$

But what does $(\mathbf{X}^\top \mathbf{X})^{-1}$ mean? Inverse of a matrix? 🤖

Matrix inverse

- The inverse of regular number x is x^{-1} which is defined by

$$x^{-1}x = xx^{-1} = \frac{x}{x} = 1$$

- **Inverse of $p \times p$ matrix $\mathbf{A}$** is denoted by $\mathbf{A}^{-1}$ and defined by

$$\mathbf{A}^{-1}\mathbf{A} = \mathbf{A}\mathbf{A}^{-1} = \mathbf{I}_p$$

where $\mathbf{I}_p$ is the $p \times p$ **identity matrix**

$$\mathbf{I}_p = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}$$

- > `A = matrix(c(2,3,3,2), 2, 2, byrow = TRUE)`
- > `invA = solve(A)`
- > `invA %*% A # returns the identity matrix`

Least squares estimate

- Least squares minimizes the sum of squared residuals

$$Q(\beta_0, \beta_1) = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- Find minimum of $Q(\beta_0, \beta_1)$ by solving system of equations

$$\frac{\partial Q(\beta_0, \beta_1)}{\partial \beta_0} = \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(-1) = 0$$

$$\frac{\partial Q(\beta_0, \beta_1)}{\partial \beta_1} = \sum_{i=1}^n 2(y_i - \beta_0 - \beta_1 x_i)(-x_i) = 0$$

gives the so called normal equations

$$n\bar{y} = n\beta_0 + \beta_1 n\bar{x}$$

$$\sum_{i=1}^n x_i y_i = \beta_0 n\bar{x} + \beta_1 \sum_{i=1}^n x_i^2$$

- With solution

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Least squares estimate - vector form

- Sum of squared residuals in vector notation

$$Q(\beta) = \sum_{i=1}^n (y_i - x_i^\top \beta)^2 = (\mathbf{y} - \mathbf{X}\beta)^\top (\mathbf{y} - \mathbf{X}\beta)$$

- Set **gradient** vector equal to zero

$$\frac{\partial}{\partial \beta} Q(\beta) = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\beta) = 0$$

gives the normal equations

$$\mathbf{X}^\top \mathbf{X} \beta = \mathbf{X}^\top \mathbf{y}$$

- Multiply both sides with the matrix inverse of $\mathbf{X}^\top \mathbf{X}$

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \beta = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

gives the least squares solution

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Gradients

- Bivariate function $z = f(x, y)$.
- **Partial derivative in x :** change in x , *holding y constant*

$$f_x(x, y) = \frac{\partial}{\partial x} f(x, y)$$

- **Partial derivative in y :** change in y , *holding x constant*

$$f_y(x, y) = \frac{\partial}{\partial y} f(x, y)$$

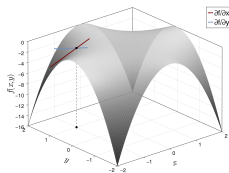
- **Gradient** is the vector of partial derivatives

$$\begin{pmatrix} f_x(x, y) \\ f_y(x, y) \end{pmatrix}$$

- General $f(x_1, \dots, x_p)$ or $f(\mathbf{x})$. **Gradient** is p -dim vector

$$\frac{\partial}{\partial \mathbf{x}} = \begin{pmatrix} \frac{\partial}{\partial x_1} f(\mathbf{x}) \\ \vdots \\ \frac{\partial}{\partial x_p} f(\mathbf{x}) \end{pmatrix}$$

Gradients



Determinant of a square matrix

- Let $\mathbf{A}$ be a 2×2 matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

- The **determinant** is the number

$$|\mathbf{A}| = a_{11}a_{22} - a_{12}a_{21}$$

- Better intuition soon on why the determinant is important.
- Formulas for larger matrices are complicated. Use a computer.

```
> A = matrix(c(2,3,3,2), 2, 2)
> det(A) # returns -5
```

Bivariate normal distribution

- X and Y follow a **bivariate normal distribution**

$$(X, Y) \sim N(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho)$$

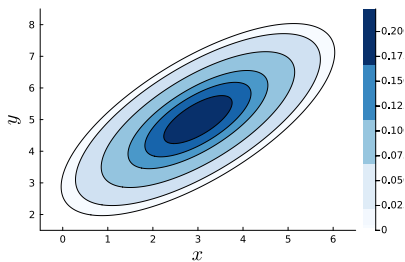
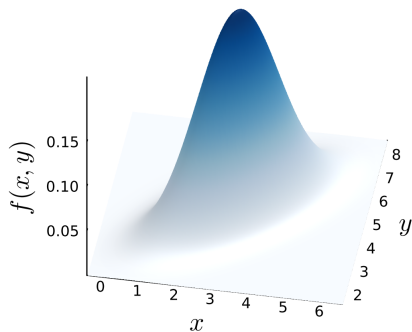
with **joint pdf**

$$f(x, y) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \times \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_x}{\sigma_x}\right)^2 + \left(\frac{y-\mu_y}{\sigma_y}\right)^2 - 2\rho\left(\frac{x-\mu_x}{\sigma_x}\right)\left(\frac{y-\mu_y}{\sigma_y}\right)\right]\right)$$

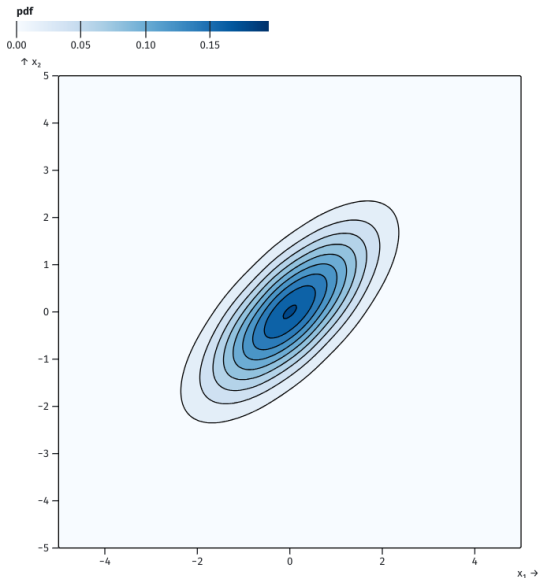
- Parameters:

- ▶ μ_x the mean of X
- ▶ μ_y the mean of Y
- ▶ σ_x the standard deviation of X
- ▶ σ_y the standard deviation of Y
- ▶ ρ the correlation between X and Y

Bivariate normal distribution



Bivariate normal - widget



Properties bivariate normal distribution

- Let X and Y follow a bivariate normal distribution

$$(X, Y) \sim N(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho)$$

- The **marginal density for X** is also normal

$$X \sim N(\mu_x, \sigma_x^2)$$

with the same parameters as those in the bivariate normal.

- The **marginal density for Y** is also normal

$$Y \sim N(\mu_y, \sigma_y^2)$$

- **Conditional densities** $f_{Y|X}(y)$ and $f_{X|Y}(x)$ are normal, see [wikipedia](https://en.wikipedia.org/wiki/Conditional_distribution_of_bivariate_normal_distribution), if you are curious.

Multivariate normal distribution

- $\mathbf{x} = (X_1, X_2, \dots, X_p)^\top$ and follows a **multivariate normal distribution**

$$\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$$

with **joint pdf**

$$f(\mathbf{x}) = \frac{1}{(2\pi)^{p/2}} |\boldsymbol{\Sigma}|^{-1/2} \exp \left(-\frac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1} (\mathbf{x} - \boldsymbol{\mu}) \right)$$

- Clash in notation: *small* bold letters for random vectors.
- Parameters when $p = 2$:

► **Mean vector**

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$$

► **Covariance matrix**

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

Multivariate normal distribution

- Determinant measures **total variance**

$$|\Sigma| = \sigma_1^2 \sigma_2^2 (1 - \rho^2)$$

- ▶ No correlation: $|\Sigma| = \sigma_1^2 \sigma_2^2$
- ▶ Strong positive correlation: $|\Sigma|$ small
- ▶ Strong negative correlation: $|\Sigma|$ small

- The quadratic form

$$(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

is the vector version of a squared standardized variable

$$\left(\frac{X - \mu}{\sigma} \right)^2$$

Linear combinations of multivariate normals

- Linear combination of random variables

$$\mathbf{a}^\top \mathbf{x} = a_1 X_1 + \dots a_p X_p$$

- If $\mathbf{x} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ then

$$\mathbf{a}^\top \mathbf{x} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$$

- Bivariate case $p = 2$.

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \quad \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \quad \boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$$

- Note

$$\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \begin{pmatrix} a_1 & a_2 \end{pmatrix} \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + 2a_1 a_2 \rho\sigma_1\sigma_2$$

$$\mathbb{V}(a_1 X_1 + a_2 X_2) = a_1^2 \mathbb{V}(X_1) + a_2^2 \mathbb{V}(X_2) + 2a_1 a_2 \text{Cov}(X_1, X_2)$$

so

$$a_1 X_1 + a_2 X_2 \sim N(a_1 \mu_1 + a_2 \mu_2, a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + 2a_1 a_2 \rho\sigma_1\sigma_2)$$